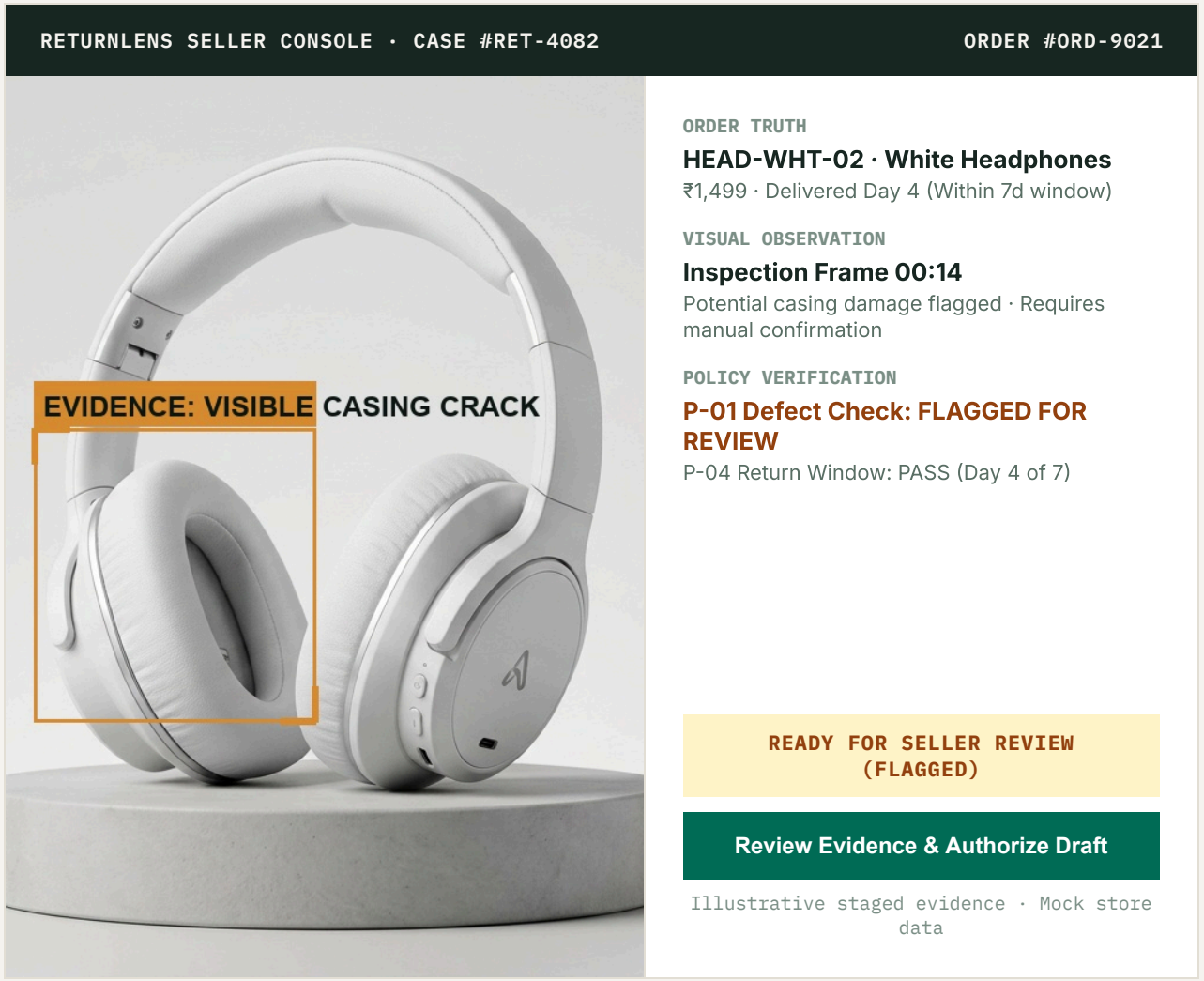
AGENTX HACKATHON 2026 · TRACK PS-04

ReturnLens

“Every return decision, backed by evidence.”

An evidence-backed return triage agent for small direct-to-consumer sellers.



Team Rookie · 3 members · PS-04 Track
Target User: Solo D2C merchant managing orders and returns alone.
Scope in 24h: Return triage core + 1 zero-vision order-status route.
Guardrails: Fail-closed architecture · No autonomous refunds.

THE PROBLEM · SOLO SELLER BOTTLENECK

One return decision. Four disconnected sources.

The return decision is made from memory, before the evidence is checked.

01 BUYER CHAT

“Received damaged item”

Buyer upload: 30–60s unboxing video + reason in support message

02 ORDER RECORD

White Headphones · ₹1,499

Shopify mock store API product inventory catalog

CRITICAL INFORMATION GAP

Four systems. No traceable decision.

NO SHARED EVIDENCE RECORD

Order facts, video evidence and return policies are disconnected — leaving one solo seller to make high-stakes refund judgments from memory.

A wrong refund loses inventory; an unfair denial loses a customer.

03 COURIER TRACKING

Delivered 4 days ago

Shippo mock tracking API scan timestamp verification

04 RETURN POLICY

7-Day Window · Sealed Tag

Deterministic code rules P-01 to P-08, never prompts

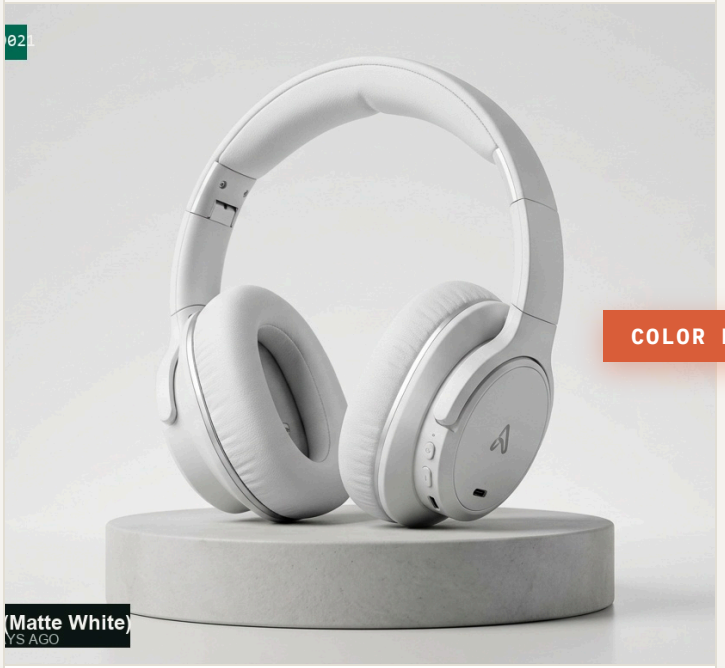
Fraud estimates differ by definition: 9% (NRF 2025, fraud only) vs 15.14% (Appriss/Deloitte 2024, fraud + abuse).

VISUAL EVIDENCE BENCHMARK · CORE DEMONSTRATION

The order says white. The video shows black.

ReturnLens surfaces the contradiction instead of auto-denying.

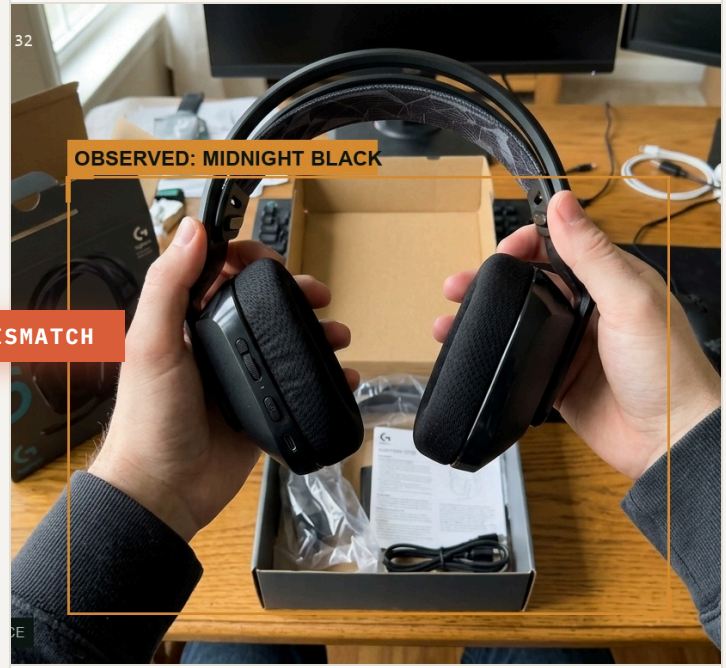
ORDER RECORD · STORE GROUND TRUTH



HEAD-WHT-02 · Matte White · ₹1,499
Delivered 4 days ago · Original retail tag present

COLOR MISMATCH

VIDEO FRAME · INTAKE TIMESTAMP 00:07



Observed Color: Midnight Black
Exact SKU: Not verified · Tag: Not visible in cited frame

DETERMINISTIC POLICY GATE

P-03 Check: COLOR CONTRADICTION

Specification contradiction flagged against Shopify order manifest.

```
typed JSON, requires_seller_approval: true
```

STATUS: ESCALATE (SELLER REVIEW REQUIRED)
Color mismatch observed · Exact SKU unconfirmed · No auto-denial.

SELLER ACTION OPTIONS

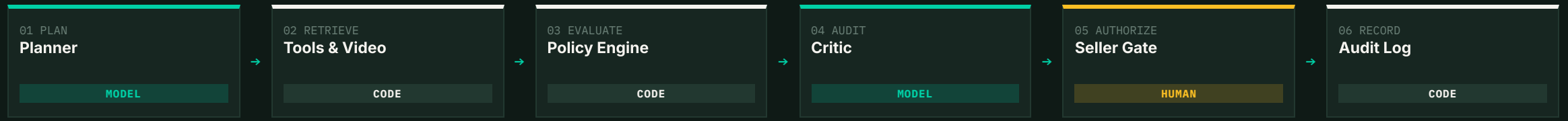
1. Accept as merchant mis-shipment (seller mistake)
2. Ask customer for outside-box shipping label (buyer mistake)
3. Hold return for physical inspection at warehouse (potential swap)

Deterministic Guardrails:
- Refund ceiling: 0 < amount ≤ min(₹4,999, remaining refundable, paid amount)
- No automatic customer denial · No refund released without seller click

SYSTEM ARCHITECTURE · DYNAMIC AGENTIC GOVERNANCE

Models observe. Code verifies. Seller authorizes.

Dynamic action selection and stateful resumption under deterministic human authorization.



DYNAMIC STATEFUL CLARIFICATION LOOP (CASE C)

Observe → Decide → Act → Wait → Resume

Dark Frames → ask_for_evidence() → WAITING_FOR_EVIDENCE → Customer Uploads → Resume Case → Seller Review

SECONDARY PS-04 ROUTE: ORDER STATUS FAST PATH

Zero-Vision Direct Route

"Where is my order?" → Query Store & Shippo APIs → Delivery Scan & Time

1. CONTRADICTION → ESCALATE

Discrepancy flagged (e.g. Case B). Inhibits automated label/refund. Seller reviews 3 options.

2. HARD FAIL → ESCALATE

Mandatory rule violation (e.g. expired return window). Seller makes final decision; never auto-deny.

3. UNCLEAR ONLY → ASK MORE

Low brightness or blurry barcode (e.g. Case C). Prompts exact re-film checklist (1 retry limit).

4. ALL PASS → APPROVE REVIEW

All clauses pass (e.g. Case A). Prepares mock label and drafts refund in pending_return_receipt.

- **Model Calls:** Bounded design estimate (3 base calls: planner, vision, critic + retries)
- **Policy Rules:** Deterministic Python code rules P-01 to P-08, never prompts
- **Critic Duty:** Reviews cited frames, flags unsupported observations (correlated error limits noted)
- **Safety Boundary:** Backend authorization boundary — LLMs cannot execute financial refund tools

DEMO SCENARIOS · GROUNDED VERIFICATION

Four scenarios. Four predictable behaviors.

Different evidence conditions produce different controlled agent actions.

CASE A · READY	**Consistent Evidence** · ORD-1042 (₹1,499) · SKU matches manifest, tag visible, defect confirmed within 7d.	→	Seller approval releases mock return label and refund draft in **pending_return_receipt**, not executed refund.
CASE B · MISMATCH	**Color Contradiction** · HEAD-WHT-02 (White) ordered vs Frame 00:07 Black item · SKU unverified.	→	**ESCALATE** (no label, no refund, no denial). Seller options: mis-shipment / box label / physical inspection.
CASE C · UNCLEAR	**Insufficient Visibility** · 4-second dark video, brightness 28, only 2 usable frames.	→	Agent requests targeted photo; enters **WAITING_FOR_EVIDENCE**; resumes case on upload (1 retry limit).
CASE D · CONTEXT	**Sequence Continuity** · Staged parcel-to-product take: opening sealed parcel to product reveal.	→	Expected behavior: inspect sequence continuity and flag missing transitions for seller review.

EXECUTION PLAN · TEAM ROOKIE (3 MEMBERS)

Built for a 24-hour sprint.

Buildable in 24 hours by 3 people, if the scope stays narrow.

00H – 06H · FOUNDATION

Mock APIs & Agent Core

Person A (Backend), supported by C
LangGraph skeleton, mock store/courier fixtures, order status route

06H – 12H · EVIDENCE

Evidence Pipeline & Policy

Person B (Video & UI)
FFmpeg keyframe indexer, Python rules P-01 to P-08, critic prompt

12H – 18H · INTEGRATION

State Flow & Scenarios

Person C (Integration), supported by A/B
End-to-end wiring, clarification/resume loop, seller console UI

18H – 24H · VALIDATION

Testing & Submission

All 3 Members
Fallback testing, authorization tests, demo video recording, README

NEVER CUT (CORE ENGINE)

Deterministic policy engine (P-01 to P-08) · Evidence-pack frame citation · Fail-closed seller approval gate · Structured JSON audit log · Clarification/resume loop · 4 demo cases.

CUT FIRST (DEFERRED EXPANSION)

Video OCR barcode scanner · Auto-drafted customer email reply · Seller analytics dashboard · Production payment gateways / live WhatsApp webhooks.

"The winning hackathon agent is not the one with the most tool definitions. It is the one where every action is held to evidence, and the seller stays in control."